Sequence in Real Analysis

August 18, 2025

Outline

Real Sequence: Definition

Definition

A mapping $f : \mathbb{N} \to \mathbb{R}$ is said to be a *sequence in* $\mathbb{R}$, or a *real sequence*.

A sequence f is generally denoted by the symbol $\{f(n)\}$. Also the symbol $\{f(1), f(2), f(3), \ldots\}$ is used to denote the sequence f.

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5. $f(n) = \sin\frac{n\pi}{2}, \quad \{1, 0, -1, 0, 1, 0, \dots\}$
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6. $f(n) = 2, \quad \{2, 2, 2, \ldots\}$ called a *constant sequence*.

Sometimes it is convenient to specify $f(1)$ and describe $f(n+1)$ in terms of $f(n)$ for all $n \geq 1$.

Definition of Bounded Sequences

A real sequence $\{f(n)\}$ is said to be:

- **Bounded above** if there exists $G \in \mathbb{R}$ such that $f(n) \leq G$ for all $n \in \mathbb{N}$.
  - G is called an *upper bound*.
- **Bounded below** if there exists $g \in \mathbb{R}$ such that $f(n) \geq g$ for all $n \in \mathbb{N}$.
  - g is called a *lower bound*.
- **Bounded sequence** if there exist $g, G \in \mathbb{R}$ such that

$$g \leq f(n) \leq G \quad \forall n \in \mathbb{N}.$$

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- In this case, the range of the sequence is a bounded set.
- For a sequence bounded above, the supremum exists: $\sup\{f(n)\}$.
- For a sequence bounded below, the infimum exists: $\inf\{f(n)\}$.

Supremum of a Sequence

The least upper bound $M = \sup\{f(n)\}$ satisfies:

(i) $f(n) \leq M$ for all $n \in \mathbb{N}$,

(ii) For each $\epsilon > 0$, there exists $k \in \mathbb{N}$ such that $f(k) > M - \epsilon$.

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If the sequence is unbounded above, we define $\sup\{f(n)\} = +\infty$.

Infimum of a Sequence

The greatest lower bound $m = \inf\{f(n)\}$ satisfies:

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5.3 Limit of a Sequence

Definition: Let $\{f(n)\}$ be a real sequence. A real number l is said to be the *limit* of $\{f(n)\}$ if:

$$|f(n) - l| < \epsilon \quad \text{for all } n \geq k,$$

for some natural number k (depending on ϵ).
Equivalently,

$$l - \epsilon < f(n) < l + \epsilon \quad \forall n \geq k.$$

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Proof (Outline):

- Assume $\{f(n)\}$ has two distinct limits $l_1 < l_2$.
- Choose $\epsilon = \frac{1}{2}(l_2 - l_1) > 0$.
- Then $(l_1 - \epsilon, l_1 + \epsilon)$ and $(l_2 - \epsilon, l_2 + \epsilon)$ are disjoint.
- For sufficiently large n, $f(n)$ must lie in both intervals — impossible.

$$\implies l_1 = l_2 \quad \text{(contradiction if different).}$$

Definition: A real sequence $\{f(n)\}$ is called *convergent* if it has a limit $l \in \mathbb{R}$.
We write

$$\lim f(n) = l.$$

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3. Constant sequence $\{2, 2, 2, \ldots\} \to 2.$

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- Then $\exists k$ such that $l - 1 < f(n) < l + 1$ for all $n \geq k$.
- Define

$$B = \max\{f(1), \ldots, f(k-1), l+1\}.$$

$$b = \min\{f(1), \ldots, f(k-1), l-1\}.$$

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Thus $\{f(n)\}$ is bounded. □

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- Corollary: An **unbounded sequence cannot converge**.
- But: A bounded sequence may not be convergent.
  Example: $\{(-1)^n\}$ is bounded but has no limit.

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5.5 Limit Theorems

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(ii) For $c \in \mathbb{R}$, $\lim(cu_n) = cu$,

(iii) $\lim(u_n v_n) = uv$,

(iv) $\lim \frac{u_n}{v_n} = \frac{u}{v}$, provided $v \neq 0$ and $v_n \neq 0$.

Proof of (i): $\lim(u_n + v_n) = u + v$

$$\begin{aligned} |(u_n + v_n) - (u + v)| &= |(u_n - u) + (v_n - v)| \\ &\leq |u_n - u| + |v_n - v|. \end{aligned}$$

Proof of (i): $\lim(u_n + v_n) = u + v$

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- Choose $\epsilon > 0$.
- Since $\{u_n\} \to u$, $\exists k_1$ such that $|u_n - u| < \frac{\epsilon}{2}$ for $n \geq k_1$.
- Since $\{v_n\} \to v$, $\exists k_2$ such that $|v_n - v| < \frac{\epsilon}{2}$ for $n \geq k_2$.
- For $n \geq \max(k_1, k_2)$, we have $|(u_n + v_n) - (u + v)| < \epsilon$.

Thus, $\lim(u_n + v_n) = u + v$. $\square$

Theorem

Let $\{u_n\}$ be a convergent sequence of real numbers converging to u. Then the sequence $\{|u_n|\}$ converges to $|u|$.

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It follows that $\big| |u_n| - |u| \big| < \epsilon$ for all $n \geq k$. Since ϵ is arbitrary, $\lim |u_n| = |u|$.

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The converse of the theorem is not true.
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Example: Let $u_n = (-1)^n$. Then $\{|u_n|\}$ converges to 1 but $\{u_n\}$ diverges.

Note

The theorem states that $\lim |u_n| = |\lim u_n|$, provided the limit on the right-hand side exists.

Theorem 5

Let $\{u_n\}$ be a convergent sequence of real numbers and there exists a natural number m such that $u_n > 0$ for all $n \geq m$. Then $\lim u_n \geq 0$.

Proof

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Proof

Let $\lim u_n = u$ and if possible let $u < 0$.
Choose $\epsilon > 0$ such that $u + \epsilon < 0$. Since $\lim u_n = u$, there exists k_1 such that

$$u - \epsilon < u_n < u + \epsilon < 0 \quad \text{for all } n \geq k_1.$$

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Choose $\epsilon > 0$ such that $u + \epsilon < 0$. Since $\lim u_n = u$, there exists k_1 such that

$$u - \epsilon < u_n < u + \epsilon < 0 \quad \text{for all } n \geq k_1.$$

Let $k = \max\{k_1, m\}$. Then $u_n > 0$ (by hypothesis) and $u_n < 0$ (from above) for all $n \geq k$. Contradiction. Therefore $\lim u_n \geq 0$.

Note

A convergent sequence of positive numbers may converge to 0.
Example: $u_n = \frac{1}{n}$, where $u_n > 0$ but $\lim u_n = 0$.

Theorem

Let $\{u_n\}$ and $\{v_n\}$ be two convergent sequences and there exists $m \in \mathbb{N}$ such that $u_n > v_n$ for all $n \geq m$. Then $\lim u_n \geq \lim v_n$.

Proof

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Proof

Let $\lim u_n = u$, $\lim v_n = v$ and $w_n = u_n - v_n$. Then $\{w_n\}$ is convergent, $w_n > 0$ for all $n \geq m$ and $\lim w_n = u - v$. By Theorem 5, $u - v \geq 0$. Hence, $\lim u_n \geq \lim v_n$.